

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Horizontal Directional Drilling (HDD) Philosophy
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Horizontal Directional Drilling (HDD) Philosophy

R01	03.06.22	Issued for Review	E. Omeke	E. Nwachukwu	K. Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	03.06.22	First Issue for Client's Review
R02		
A01		

Review Code & Status (to be filled by Client)		
☐ Code 1	Approved – Further work may proceed.	
☐ Code 2	Approved with comments. – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ Code 3	Reviewed – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ Code 4	Retained for information only	
☐ Code 5	Review Not Required – Do not Resubmit	
Date:	Position:	Sign:

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- PPL-PHY-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Wall Thickness Calculation Report
EA-BFPCL-GPMC-GGPL-FEED-PPL-ALS-001 to 037	Pipeline Alignment Sheet
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010	Pipeline Cleaning, Gauging, & Hydrotesting Specification.
EA-BFPCL-GPMC-GGPL-FEED-PPL-PHY-001	Pipeline Integrity Management Philosophy.
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009.	Specification for Pipeline Construction.

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

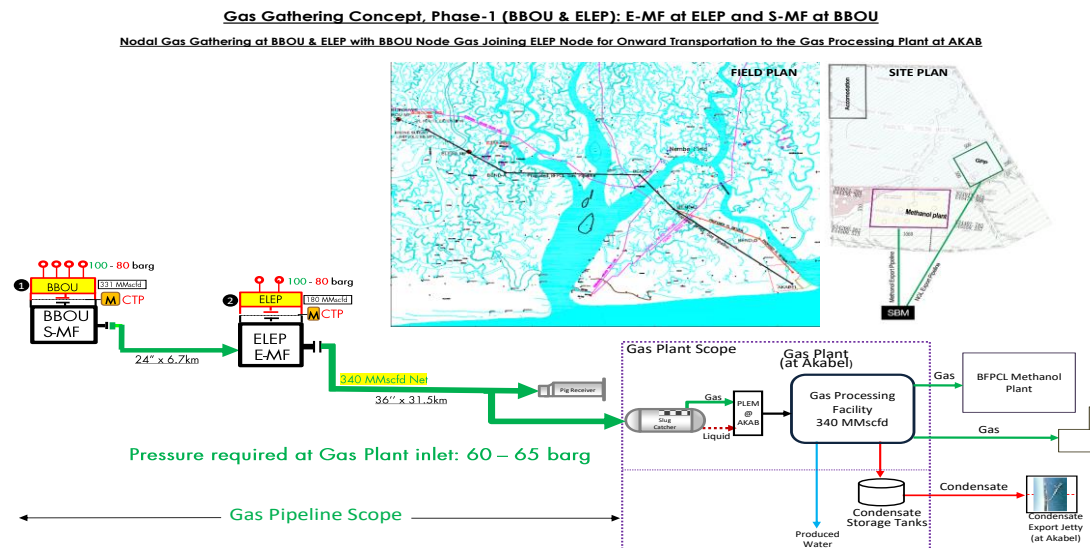


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document specifies the general requirement for the installation of onshore Pipelines using the horizontal directional drilling (HDD) techniques. It shall be read in conjunction with the document Specification for Pipeline Construction - EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009. Any requirement mentioned therein with regards to Site and ROW preparation not fully elaborated in this document shall apply.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP. 31.40.20.37-Gen	Line pipe for Critical Service (amendments/supplements to IOGP S-616 AND DNVGL-ST-F101 2019)
DEP. 82.00.10.10-Gen	Project Quality Assurance
DEP. 31.40.40.38-Gen	Hydrostatic Pressure Testing of new Pipeline (February 2017)
DEP. 61.40.20.30-Gen	Welding of Pipelines & Related Facilities (Amendments / supplements to API Std 1104)
ASME B31.8	Gas Transmission and Distribution Piping Systems
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
API 5L	Specification for Line Pipe
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
API RP 5LW	Recommended Practice for Transportation of Line pipe on Barges and Marine Vessels
API RP 5L1	Recommended Practice for Railroad Transportation of Line Pipe
API 1104	Standard for Welding Pipelines and Related Facilities
API RP 5L6	Transportation of Line Pipe on Inland Waterways
ASNT SNT-TC-1A	Personnel Qualification and Certification in Non-Destructive Testing
API 1102	Steel Pipeline Crossing Railroad and Highways
ISO 3183: 2007	Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems.

ISO 9001	Quality Systems Model for Quality Assurance in Design, Development, Production, Installation and Servicing
ISO 13623	Petroleum and Natural Gas Industries - Pipeline Transportation Systems
ISO 3183	Petroleum and Natural Gas Industries – Steel Pipe for Pipeline Transportation Systems
ISO TS 29001	Petroleum, Petrochemical and Natural Gas Industries – Sector Specific Quality Management Systems Requirements for Product and Service Supply Organizations

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Line Pipe Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Table 3.1: 24" Multiphase Pipeline Data

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	24"
Density	7,850 kg/m ³
Young's Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570 MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
OD / Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	40mm (to be confirmed)
Installation Condition	Trenching
Cathodic Protection	Impressed current

3.1. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

Table 3.2: 36" Multiphase Pipeline Data

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	36"
Density	7,850 kg/m ³
Young's Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
OD / Wall Thickness	23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	40mm (to be confirmed)
Installation Condition	Trenching
Cathodic Protection	Impressed current

4. Environmental and Pipeline Design Data

Refer to the document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 - Basis of Design

5. General Requirements for HDD

5.1. Approvals

The following authorities may be required to authorize the works: CONTRACTOR shall be advised on the appropriate authority to apply for approval prior to the works.

- Federal ministry of Transport.
- Federal Ministry of Petroleum.
- Department of Petroleum Resources (DPR).
- Nigerian Maritime Administration and Safety Agency (NIMASA).
- State Government.
- Local Government.
- Other Service Providers.
- Private Landowners.

No work is to begin on site preparation or HDD activities until all relevant permits and approvals have been gained and signed off by the relevant authority.

The CONTRACTOR shall adhere to any approval conditions that are specified in the authorizations.

5.2. HDD Site Survey

Prior to the performance of the drilling operation, the CONTRACTOR shall perform a geotechnical investigation and borings to confirm the subsurface conditions in the crossing area. CONTRACTOR shall perform a minimum of two borings on each side of the river (or Crossing) and preferably one boring in the middle of the river, with an offset of minimum 15 m to the projected pipeline Centerline. Borings shall be performed to a depth 9 m below the maximum crossing depth.

The location, coordinates, and elevation of survey control points in the vicinity of the work site will be provided together with the following drawings.

- horizontal pipeline alignment.
- location of the soil borings.
- approximate locations of existing and planned substructures.
- location of existing surface structures.
- location of existing property boundaries.

- Preliminary HDD profile.
- elevation of existing substructures and other civil works.
- work areas for the drill site.
- pipe string fabrication area.

5.3. HDD DESIGN

The CONTRACTOR shall be responsible for the design of all aspects of the HDD works including any temporary works and temporary supporting structures. All design assumptions regarding sub-surface conditions, equipment requirements, groundwater and other factors are the responsibility of the CONTRACTOR and shall be fully documented as follows.

- Drilling fluid program.
- Rig selection analysis.
- Tooling and drill pipe selection
- Penetration rates.
- Conductor casing selection and sizing
- Casing pipe.
- Hydro-fracture risk study.
- Time and Cost analysis.
- Potential pullback loads.
- Carrier pipe selection.
- Temporary works design

Based on the alignment shown in the profile Drawings, the CONTRACTOR shall design and size the excavated profile to accommodate all temporary and permanent works.

A detailed design plan view and profile drawings shall be submitted to the EMPLOYER for review prior to commencement of work.

The CONTRACTOR shall not proceed with any work until EMPLOYER approval on all engineering design documentation.

The HDD crossing shall be designed in accordance with this Specification and the referenced documents.

The profile design shall consider the following:

- Temporary works associated with the HDD construction.

- Pipe/Drill rod bending radii.
- Steering capability of the proposed method/equipment.
- Drilling fluid performance.
- Tensile loads (expected and maximum allowable).
- Potential for hydro-fracture.
- Pipeline materials properties (typically Steel,) & operations requirements (lifespan and loads).
- Location of existing services.

The CONTRACTOR shall be responsible for submitting a design document package for review and approval by EMPLOYER. The document package shall form part of the CONTRACTOR's ITP as a hold point. The document package shall include but not limited to:

- Detailed profile design drawings.
- Temporary works design – Drill rig thrust restraints, shoring systems, and pits.
- Drilling Fluid Design and Management Program.
- Settlement.

The Profile Drawings shall be used as a guide to confirm that the crossing is possible to be delivered via HDD methods. The CONTRACTOR is not to rely on the Profile Drawings as being the built solution as it is the CONTRACTOR's responsibility for the final alignment and design, and any temporary works design required.

The EMPLOYER shall be responsible for the operation of the permanent pipeline; however, the CONTRACTOR is required to confirm the suitability of this pipe for installation via HDD methods.

Under the design requirements of this Specification the CONTRACTOR is to produce the Drawings outlined in Table 6 below.

Table 5.2: Drawing Requirements

Drawing Details	Plan/ Elevation	RPE Sign Off
Site Layouts (incl. Laydown, Storage areas, Entry and Exits)	Plan	
HDD Alignment	Plan + Elevation	
Borehole details	Elevation	
HDD Pipe Stringing Layout	Plan	

HDD Over Bend Details	Plan + Elevation	
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5.3.1. Pipe Stress Analysis

CONTRACTOR shall carryout stress checks line pipe viz a vis the profile of the HDD. This will include the analysis of the combined stresses to be experienced by the pipe during operation (e.g., internal pressure, external surcharge loads, temperature, seismic induced ground shaking, ground faulting, bending, etc.).

This analysis may include the stresses to be imposed during installation.

5.3.2. Required Documentation

As a minimum and in addition to the documentation required in the CONTRACTOR Management Plan Requirements, the following documents and calculations listed in Table below shall be submitted by the CONTRACTOR for EMPLOYER's review and approval.

Table 5.2.2: Required Documentation

Design Documentation Submittals	Submission (Minimum)
Detailed Profile Design Drawings	4 weeks before work
Temporary Works Design – Drill Rig Thrust restraint, shoring systems, pits etc.	4 weeks before work
Drill Fluid Design and Management Programme	4 weeks before work
Drilling analysis	4 weeks before work
Drilling rig anchorage design	4 weeks before work
Pipeline and Conduit installation stress analysis	4 weeks before work
Hydro-fracture (Frac-out) Analysis	4 weeks before work
Pipeline and Conduit pulling head design	4 weeks before work
Any other temporary works designs	3 weeks before work

5.3.3. Drilling Analysis

The Drilling Analysis shall include, but not be limited to, the following:

- Drilling profile design;
- HDD equipment selection;

- Downhole tooling selection;
- Drill string analysis to confirm suitability of tooling;
- Maximum required pull force calculation;
- Pipeline ballasting design;
- Spoil separation system assessment;
- Drilling fluid design;
- Ground settlement analysis;
- Drilling fluid hydro-fracture analysis.

5.4. Communications

CONTRACTOR shall provide personnel with communication devices to facilitate communication between the crew members on site and in the office during the drilling operations and to report any drilling fluid releases (frac-out) along the full length of the horizontal directional drill (HDD) corridor at all times.

5.5. Site Security

CONTRACTOR shall install EMPLOYER reviewed and accepted safety warnings, lights, and/or signage along and around all work areas and/or pipe stringing areas to provide appropriate notice to the general public of the activities being carried out and to prevent personal harm or damage.

5.6. Hammer Devices

CONTRACTOR shall not use any hammering device for installation purposes without prior EMPLOYER review and acceptance.

5.7. Drilling Plan

CONTRACTOR shall submit a drilling plan for EMPLOYER review and acceptance at least 45 days prior to beginning work.

Acceptance of drilling plan by EMPLOYER does not relieve CONTRACTOR of any responsibility or liability for safety, damages, compliance with permits and regulations, accuracy, or adequacy of the plan for execution of the project.

The drilling plan shall be site specific and shall be updated during drilling operations.

The plan shall consider the anticipated drilling conditions, soil conditions, depth of groundwater, and other site-specific parameters in accordance with the geotechnical boring information

provided by the EMPLOYER. CONTRACTOR's drilling plan shall not override or change the requirements of Project Drawings, Specifications, Contract Documents or this specification without review and written acceptance from EMPLOYER.

The following items shall be included in the plan or pack which shall be submitted to EMPLOYER 4 week before the commencement of HDD activities.

- Drilling rig(s) description, including the manufacture and type of rig, size of drilling rig(s)
- Major Lift Plan(s)
- including torque and pulling/pushing capacities.
- Details and description of ancillary equipment required at the HDD entry and exit locations and space requirements.
- Details of any major maintenance or repair work reasonably expected to be performed on the HDD rig or associated equipment and how it will be performed safely (e.g.: access platform, lifting equip, power tools, etc....)
- Type of tracking equipment to be utilized during the pilot hole process, its stated accuracy and sketch of tracking cable layout. This shall also include a description of the down hole survey instruments and surface location equipment.
- Description of drilling operation, including pilot hole size, each reaming step size, and final hole size.
- Diameter, type, grade of drill pipe.
- Manufacturer and type of pilot hole bits, including diameter.
- Manufacturer and type of hole opening tools, including diameter(s).
- Anticipated and maximum penetration rates to be maintained during each phase of the operation.
- Description of the fresh water system, including the equipment required to obtain and store fresh water, anticipated source for gathering fresh water, anticipated quantity of water required for drilling operations with estimated maximum daily rate to be consumed.
- Description of the drilling fluid system, including mixing tank(s) volume, cleaning tank(s)
- volume, mud pump(s) capabilities (pressures, flow rates, etc.), recycling plans.
- Description of the action to be taken in the event that any obstructions are encountered during drilling operations.
- Description of surface settlement monitoring, if required by permit and/or landowners requirements (e.g., borehole extensometers), for HDD crossings in the vicinity of surface

structures and other features sensitive to ground settlement (e.g., rail lines, roadways, buildings, etc.). Contingency measures to be employed should surface settlement be experienced during HDD operations shall also be discussed.

- Anticipated drilling fluid composition, bulk unit weight, and additives to be used, including all Material Safety Data Sheets.
- Description of the disposal method to be used for disposing of drilling fluids and hole cuttings, including a list of any SUB-CONTRACTORS to be used in the handling or transportation of drilling fluids, the frequency and volume of the anticipated disposal, and the disposal location.
- Contingency plan for handling inadvertent drilling fluid migration and release (frac-out), including an inventory of the equipment and material to be on-site for mitigating the impacts of fluid release.
- The contingency plan shall be site specific and shall consider any sensitive environments which may be impacted by HDD operations.
- Calibration certificates shall be provided for all instruments requiring calibration, as identified herein (e.g., drilling fluid monitoring equipment)
- Description of the materials and equipment to be used to clear the line of debris and ensure the integrity of the installed pipe string (e.g., cleaning pigs and/or sizing plates)

5.8. Daily Reports

CONTRACTOR shall submit daily reports to EMPLOYER at the end of each work shift.

The daily reports shall include the following information:

- The names of the supervisor, crew members, third party, and contract personnel on site;
- The beginning and ending times of the shift;
- Description of the work performed, tools in use, and footage completed.

Daily total of drilling mud, additives, and other materials used and the total quantities used to date;

- Quantity of drilling fluid and cuttings lost, released, and/or hauled from drill site.
- Description of any pilot drill deviations and corrections made.
- The typical and maximum torques, and push pull forces used, including an explanation for any excessive values beyond normal operating ranges.
- Penetration rates for each pilot or reaming effort.

- As-built pilot hole survey information (horizontal and vertical alignment), including a description and explanation for any deviations from construction drawings.
- An update of any changes to the drilling schedule or drilling plan.

5.9. Final Reporting

CONTRACTOR shall furnish to EMPLOYER a copy of the complete computer printout of the directional survey report and an as-built map of the horizontally drilled section showing the X, Y, and Z-coordinates of the final location of the pipeline. The maximum spacing between coordinate points shall be every joint of drill pipe.

5.10. As-Built Drawings

CONTRACTOR shall furnish the as-built drawing(s) within 15 days after the carrier pipe has been installed. These drawings shall be prepared at an appropriate scale, in both the horizontal and vertical planes, unless otherwise directed by EMPLOYER.

5.11. Hydrostatic Test Records

CONTRACTOR shall furnish the hydrostatic test records within 15 days after the carrier pipe has been successfully tested.

The report shall include:

- Charts from the temperature and pressure recorders,
- Dead weight tester calibration and certification records,
- Location and elevation of dead weight tester and other instruments,
- Test medium,
- Calibration records for temperature and pressure gauges and measurement devices,
- The volumes of water drained or added to the line during the test,
- Spreadsheet of the recorded dead weight tester pressure and temperature readings, atmospheric temperature and condition (e.g., sunny, overcast, night, etc.) at thirty-minute intervals,
- A sketch of the pipe being tested, including the length, profile, beginning and ending stations, etc., and
- Any agency or third-party certification records that may be required as per the contract.

5.12. Radiographic Inspection Records

If CONTRACTOR is responsible for radiographic or other weld inspection, all inspection records

shall be furnished to EMPLOYER for review and acceptance prior to the carrier pipe being installed.

The carrier pipe shall not be installed until CONTRACTOR has received EMPLOYER acceptance of the weld inspection records.

5.13. Drilling Fluid System

CONTRACTOR shall provide a qualified and trained drilling mud technician as part of the daily work crew to assess and implement necessary adjustments to the drilling fluid composition.

A certified mud engineer shall be available to the CONTRACTOR, if requested by the EMPLOYER, to assist in the resolution of problems that may arise during the HDD operation.

Any additives not included in the EMPLOYER accepted drilling plan will NOT be allowed on-site or to be used without prior EMPLOYER review and acceptance

CONTRACTOR shall provide and maintain instrumentation to accurately measure the drilling fluid discharge rate and pressure (e.g., flow meter or stroke counter, etc.).

The instruments shall be calibrated, as applicable, within the past six months.

Calibration certificates shall be furnished to EMPLOYER prior to beginning work.

EMPLOYER shall always have access to all instruments and their readings.

CONTRACTOR shall take all necessary precautions to ensure that the drilling fluid pressure inside the drill stem (from the pressure jetting to the drilled hole) does not exceed what can be contained by the overburden soil to prevent frac-out. This shall include, but is not limited to, a thorough review of the soil conditions at the site prior to beginning work, pressure monitoring in the control, and limiting drilling fluid pressures during the HDD operation, etc.

Down hole pressure monitoring may not be required unless specified by the EMPLOYER as a special criterion.

CONTRACTOR shall make every effort to maintain circulation and recycle the drilling fluid throughout the drilling process.

Based on the soil condition noted in the soil borings, CONTRACTOR shall identify the need for, and implement the use of, casing where necessary to contain drilling fluid.

CONTRACTOR shall insure that all drilling fluids are contained within the drilled hole, storage system, and the fluid system.

Drilling fluid and cutting returns shall be reused where applicable to minimize waste.

CONTRACTOR shall remove all spent drilling fluids and associated cuttings from the job site and transport them to a EMPLOYER reviewed and accepted facility for proper disposal in accordance

with the EMPLOYER accepted drilling plan.

CONTRACTOR shall have adequate spill containment measures, including collection materials and equipment on site at all times to contain and collect any release of drilling fluids to the sensitive resources, including, but not limited to wetlands or waterways.

Any area affected by drilling fluid migration and release (frac-out) shall be cleaned-up and restored to EMPLOYER's satisfaction. The clean-up and restoration shall be performed in compliance with all project permit requirements

5.14. Pilot Hole Drilling, Tracking, and Survey

The pilot hole shall be drilled along the path shown on the project drawings.

The position of the drill string shall be monitored by CONTRACTOR with surface location equipment (e.g., TruTracker® or EMPLOYER reviewed and accepted alternative).

Precise down hole survey may not be needed unless requested by the EMPLOYER.

5.15. Pilot Drill Entry and Exit Angles Tolerance

Pilot drill entry and exit angles Tolerance shall be in accordance with Table 1 below unless noted otherwise by EMPLOYER.

Table 5.15: HDD Drilling Tolerances

Parameter	Tolerance
Pilot Drill Entry Angle	± one degree (1°)
Pilot Drill Entry Location	± 0.305m measured parallel to pipeline, ± 0.153m measured perpendicular to pipeline.
Pilot Drill Exit Angle	Plus or minus one degree (1°)
Pilot Drill Exit Location	± 0.610m measured parallel to pipeline, ± 0.153m measured perpendicular to pipeline.
Pilot Drill Vertical Elevation	+ 0mm above the stated elevation, - twenty feet 6.096m below the stated elevation.
Pilot Drill Horizontal Alignment	± 0.610m

5.16. Indemnification and Crossing of Existing Facilities

CONTRACTOR shall take all necessary precautions to protect known and unknown existing facilities. This shall include, but is not limited to the following:

- Contacting the local underground service alert agency, utility locating, or notification group

in accordance with local laws and standard practices,

- Conducting an on-site preconstruction meeting with all EMPLOYERs and operators of such facilities,
- Positively locating and staking all existing lines, cables, and other underground facilities located within the construction corridor,
- Modifying HDD design and down hole assemblies as may be necessary to prevent damage to existing facilities.

5.17. Pilot Drill Radius

The pilot hole shall be drilled in accordance with the project drawings.

In no case shall the minimum pipe radius be less than one thousand (1,000) times the nominal pipe diameter. (Generally, the design radius reflected on the project drawings reflects the combined pipe operating and installation stresses; the design radius shall be greater than the minimum allowable radius to allow for construction tolerances.)

5.18. Pilot Hole Corrections

In the event the pilot hole is outside the tolerances stated in table 1 above, CONTRACTOR shall pull-back and re-drill the pilot hole to correct the horizontal alignment and vertical profile.

CONTRACTOR shall provide documentation demonstrating that the re-drilled pilot hole is within tolerance.

CONTRACTOR will be responsible for re-drilling of any portion of the pilot hole to meet the stated tolerances, including grouting, if required, of the old pilot hole.

5.19. Hole Reaming and Preparation for Pipe Pullback

CONTRACTOR shall meet with EMPLOYER and review the pilot hole completion data before the hole-opening process begins.

CONTRACTOR may not proceed with hole-opening until EMPLOYER acceptance has been granted.

5.20. Lost or Lodged Tools

Any tools or other metal object lost or lodged down hole shall be reported to the EMPLOYER.

Metal objects shall be fully recovered prior to pipe pullback operation unless specifically accepted otherwise by EMPLOYER.

Failure to recover metal objects lost or lodged down hole within a reasonable time period

constitutes just cause for rejection of the drilled hole.

5.21. Swabbing

CONTRACTOR shall run a final swab pass prior to pull-back operations unless otherwise accepted by EMPLOYER.

5.22. Pipe Fabrication and Pullback

5.22.1. Pipe String Fabrication

Pipe string fabrication and field joint coating shall in accordance with the project specifications and drawings.

5.22.2. Hydrostatic Testing

CONTRACTOR shall perform a 4-hour hydrostatic test of the pipe string(s) at the EMPLOYER specified test pressure. This test shall be performed after all welds have been accepted by EMPLOYER, prior to commencing pullback operations.

5.22.3. Pipe Pullback

CONTRACTOR shall provide and maintain instrumentation that will accurately measure drill string axial and torsional loads. EMPLOYER ' Representative shall have access to these instruments and their readings at all times.

The maximum permissible tensile load to be imposed on the pull section shall be calculated using the following formula:

$$P = 0.8 \times SMY \times A$$

Where:

P = Maximum pull or tensile load (pounds)

SMYS = Specified minimum yield strength of pipe (psi)

A = Cross sectional area of pipe

If more than one pipe grade and/or wall thickness value is involved for a given pull section, the maximum permissible tensile load shall be calculated for each combination of wall thickness and pipe grade. The minimum value shall be used as the maximum pulling force for the pipe string.

In no event shall the pull force exceed one million (1,000,000) pounds without prior EMPLOYER review and acceptance.

5.22.4. Swivel

To minimize torsional stress imposed on the pull section, CONTRACTOR shall use a swivel

assembly to connect the pull section.

5.22.5. Continuous Pull Operation

CONTRACTOR shall install the pipeline in one continuous string unless stated otherwise in EMPLOYER accepted drilling plan.

CONTRACTOR shall begin the pullback operation immediately after the reaming and/or final swab pass has been completed.

Once CONTRACTOR begins the pullback operation, the operation shall not cease until pullback operations are completed, unless otherwise reviewed and accepted by EMPLOYER.

5.22.6. Buoyancy

CONTRACTOR shall provide buoyancy modification as required and/or when conditions necessitate.

5.23. Pipe Protection

During the pull-back operation, CONTRACTOR shall monitor pipe handling operations and shall use all required materials and equipment to assist with the safe and proper movement of the pipe. This shall include, but is not limited to, pipe rollers, side-booms, padded shoes, cushions, etc. Situations that cause external coating or other damage to the pipe string shall be corrected immediately.

5.24. Pipe Inspection

The external pipe coating and field joint coating shall be inspected in accordance with the project specific coating requirements prior to and during the pullback operation.

The pipe string shall be continuously inspected for coating damage using a holiday detector (approved by the OWNER) during the entire pull-back operation.

EMPLOYER reserves the right to stop pullback operation if coating or other damage cannot be corrected to EMPLOYER's satisfaction.

If multiple pipe strings are used, the tie-in welds between successive pipe strings shall be radiographically inspected and coated in accordance project coating requirement prior to being pulled into the HDD installation.

5.25. Lodged Pipe String

Should the pipe string become lodged in the drilled hole during pullback, CONTRACTOR shall consult with EMPLOYER.

5.26. Voids

Should it become evident that excessive materials or spoils have been removed from the annulus during reaming operations, resulting in voids around installed pipe, CONTRACTOR shall install EMPLOYER reviewed and accepted grout or other material to fill the voids.

5.27. Post Installation Inspection, Testing, Commissioning and Acceptance

5.27.1. Cleaning

Immediately after pull-back operations, the exposed ends of the installed pipe shall be cleaned to facilitate an inspection of the pipe and external coating.

The internal surface of the pipe shall be cleaned to EMPLOYER satisfaction by running a series of scraper pigs. Brush pigs shall be run to remove welding slag to protect downstream strainers.

The final tie ins to the existing pipeline shall not start until the EMPLOYER has reviewed and accepted the cleaning operations.

5.27.2. Sizing Plate

After pipe cleaning, CONTRACTOR shall run a pig with a sizing plate. The sizing plate shall be fabricated in accordance with the document EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009- Specification for Pipeline Construction to check for ovality, buckles and/or dents.

The sizing plate run is required prior to EMPLOYER acceptance and acceptance of HDD installation.

The sizing plate run will be considered acceptable if results indicate no anomalies (e.g., ovality, dents, buckles, or internal obstructions).

5.27.3. Hydrostatic Test before Installation

CONTRACTOR shall perform an 8-hour hydrostatic test of the welded string of pipe to be installed at the specified test pressures of 1.25 X DP in compliance with project requirement for mainline Hydrostatic test before installation.

5.27.4. Internal Inspection

EMPLOYER reserves the right to internally inspect the HDD installation prior to tie-ins using an internal inspection device (smart pig).

Should EMPLOYER elect to run an internal inspection device, CONTRACTOR shall be reimbursed for any additional costs that may be incurred, unless the need to run such a device is a result of the CONTRACTOR's workmanship or construction defect.

5.27.5. Hydrostatic Testing After installation

Hydrostatic testing of the carrier pipe shall be conducted after insertion of the pipe into the bore-hole.

The Carrier pipe shall be hydrostatically tested along with the adjoining pipeline segments as per Specification for Line Pipe and Specification for Pipeline Construction. The Carrier pipe shall be hydrostatically tested to $1.25 \times$ Design pressure, as per requirements in accordance with ASME B31.8. The resulting hoop stress shall not exceed 95% of SMYS of the Carrier pipe material.

The CONTRACTOR shall engage a certified testing authority to conduct the hydrostatic testing.

The CONTRACTOR's Hydrostatic Testing Procedure for the disposal of test water shall be in accordance with the Federal Ministry of Environment and Petroleum Resources Guideline.

5.27.6. Leakage Testing

For specific leakage testing requirements refer to the requirements identified in the specific Scope of Works. This may include vacuum testing.

5.27.7. Disinfection/Chlorination

The disinfection/chlorination testing reports are to be prepared by a Federal Ministry of Environment and the Department of Petroleum Resources (DPR) certified testing authority. Equipment used in the testing process shall be calibrated in accordance with the relevant standards.

The CONTRACTOR's Chlorination Testing Procedure for disposal of disinfected water shall be in accordance with the Federal Ministry of Environment, Water Resources and the Department of Petroleum Resources, the state Government and Local Government Environmental standards.

5.28. HDD Installation Acceptance

EMPLOYER will review the results of the hydrostatic test data, sizing plate run data, and internal inspection (if required) and then determine the acceptability of the installed HDD crossing.

EMPLOYER shall notify the CONTRACTOR of EMPLOYER's acceptance or rejection of the HDD installation after successful completion of the hydrostatic test.

5.29. Site Clean Up And Restoration

5.29.1. Abandoned Holes

CONTRACTOR shall fill any abandoned drill holes with grout or other EMPLOYER reviewed and accepted material to completely seal and fill the upper thirty (30) feet of the hole. In some situations, EMPLOYER material may require more extensive grouting up to and including grouting of the entire abandoned pilot hole(s) to prevent frac-out of adjacent HDD paths, or to comply with permit or other job conditions.

5.29.2. Clean Up and Restoration

CONTRACTOR shall demobilize and remove all equipment, materials, and temporary facilities.

Unless otherwise noted on the project drawings, the site shall be restored to the original lines and grades that existing prior to beginning work.

Any drilling or mud pits and/or excavations shall be backfilled by CONTRACTOR in accordance with project specification.

Final restoration shall be in accordance with applicable permits and Right-of-Way agreements.

6. HDD Quality Control (QC)

Quality Assurance (QA) and Quality Control (QC) measures should be a primary goal in order to ensure a successful HDD project, not only in terms of a successfully installed pipeline but also in terms of safety, schedule and within planned costs.

It is the duty of the CONTRACTOR to provide a site-based Inspector to monitor and review the quality of all factors involved in HDD construction process. The QC process should include and not limited to monitoring and inspection of:

- Drilling Program
- Drilling equipment
- Drilling fluid products
- Down hole assembly
- Drilling practices, and
- Site Management.

7. Personnel

Appropriately trained and experienced personnel are required for the delivery of the works. Table 13 below summarizes the minimum training and experience required for key personnel. Details of key personal experience shall be provided to the EMPLOYER's Representative for approval before the works commence.

A HDD supervisor who is thoroughly knowledgeable of the equipment, drilling and HDD procedures shall be present at the job site during the entire installation and be available to address immediate concerns and health and safety issues. As a minimum the following person with shall be engaged in the works.

Table 7: Key HDD Personnel and Experience.

HDD Role	Qualification	Experience in Role (Years)
Project Manager	Min Higher National Diploma	15
HDD Supervisor	Rig and fluid training	10
HDD Driller	Rig and fluid training	8
HDD Offsider	Fluid and Bore Tracking training	5
HDD Engineer	Min Higher Education Diploma	10
HDD Steerer	Survey equipment Training	2
HDD Mudman	Rig and fluid training	3

8. Health and Safety Requirements in HDD.

Comprehensively and thoroughly considered shall be given to Health and Safety Requirements in the planning HDD Design and Installation.

All valid national safety and accident-prevention regulations, and Environmental regulations and standards such as EGASPIN shall be adhered to.

The dangers and risks listed below are supplemental to any other valid regulation and can in no way replace them.

Prior to the start of works, the entire personnel employed on the construction site are to be informed in a briefing about the danger of accidents and preventative measures together with all local rescue measures and organizations. The first-aid attendant, the responsible safety officer and the safety specialists must be presented to all employees. The following risks of the horizontal directional drilling are to be avoided by suitable preventative measures:

- Working on inclined surfaces: The working surfaces on the machines employed must be fabricated from particularly slip preventing and easy to clean materials. Sturdy railings must prevent personnel from falling
- Working near rotating tools and machine components: The contact with rotating machine parts must be prevented as far as is possible by way of permanently installed protective guards. The work-clothes of operating personnel should fit tightly. Particular attention is to be paid to tight and closed arm-, body- and waist bands.
- Increased danger of slipping due to drilling mud: When releasing the drill pipes, bentonite-suspension which flows out of the drill pipes should be conducted into collection basins and channels. Rinsing water for cleaning the working areas must be kept constantly available on the rig, equipment and at the mixing plants.
- Dangerous strain on the respiratory tracts due to bentonite-dust: Working under bentonite-dust is to be avoided as far as is technically possible and only permissible with dust masks (particle-filtering half-masks) and close-fitting protective goggles.
- Handling of suspended loads (drill pipes, reamer, etc.): The handling of drill pipes and other loads with lifting devices is to be carried out with great care. Pipe clamps, lifting ropes and belts are to be regularly inspected for soundness.
- The feeding of drill pipes to the drill string should be carried out by stationary equipment, even at the drilling exit area. Standing under suspended loads is to be strictly avoided. An adequate safety distance from overhead power lines of all kinds must in all cases be observed and the strict procedures required by the national authorities should be adhered to at all times.

- Great torque when making up or breaking out drill string connections: Special care is to be given to observe correct and operationally safe methods for the making up and breaking out devices. In particular great care and attention should be observed when installing manual clamping collets and only experienced skilled personnel should be used. Special care is also to be taken regarding correct and safe working conditions at the drilling exit area.

9. Environmental Requirements

9.1. Environmental protection

When subsoil conditions are sensitive on the construction site, the access roads and working areas are to be made safe for heavy construction equipment and vehicles. Use of excavator mats is recommended.

When excavating the ground for the necessary pits, attention is to be paid to the type of topsoil and each additional soil layer. Separate storing is necessary and when backfilling, replacements of the original sequence of soil layers has to be followed.

It is necessary to avoid ground contamination due to spillage of oils, fuels or greases in all cases (use of oil, foils etc.). Provision should be made on the construction site for adequate oil-binding agent's storage for unforeseeable spillage occurrences.

Particular attention is to be given to safe storage and handling of drill fluid. Adequate (storage) pits have to be prepared in advance while at all times uncontrolled escape of drill fluid is to be prevented by all means.

The final disposal of residual fluid and cuttings is to be permitted by the local environmental authorities prior to start-up of the project.

10. Risk Assessment and Control

The CONTRACTOR is to prepare and implement an approved contingency plan dealing with the key HDD risks identified in the risk register. As a minimum the CONTRACTOR is to have defined plans complete with equipment and materials on standby to mitigate against the following HDD risks:

- Fluid loss.
- A hydro-fracture event.
- Hydro-lock (loss of fluid circulation).
- Hole collapse.
- Fluid pit overflow.
- Hydrocarbon spill.
- Drill pipe or Bottom Hole Assembly failure.
- Serious workplace safety incidents in remote areas.

CONTRACTOR shall provide a detail procedure for fluid management to be followed and timely notification given to EMPLOYER.

Risk assessment for project and controls shall be assessed per every drill program while ensuring that risks are effectively communicated to personnel performing the works.

11. Procurement

11.1.1. Approved CONTRACTORS

The CONTRACTOR shall provide materials which are previously tested and accepted in accordance to EMPLOYER and international codes/standards

If the CONTRACTOR proposes to utilize non pre-approved products, these are to be submitted to the EMPLOYER for consideration.

11.1.2. Delivery, Storage and Security of Materials

The CONTRACTOR shall submit to the EMPLOYER for review and approval, information detailing the proposed method of handling, stacking during storage, and stacking and securing line pipe for shipment. The stacking information shall include the maximum permissible stacking height for the line pipes, together with the calculation, (for both coated and uncoated pipes).

The CONTRACTOR shall also detail storage requirements for short (6 months) and long-term storage including any preservation and storage procedures that may be required.

No welding of temporary attachments for handling, stacking or securing shall be permitted. All handling, loading and unloading and storage shall be performed in such a manner as to avoid mechanical damage and corrosion using slings or pipe-shaped padded hooks with a spreader bar attached.

The delivery and storage of pipe materials shall be in accordance to technical specification for line pipe.

The CONTRACTOR shall provide security for the Site and Works including the construction facilities, plant and equipment. Materials shall also be secured by the CONTRACTOR to prevent their removal by unauthorized personnel.

All pipes are to be maintained clean and dry (capped, plugged or blank flanged) from manufacturers facilities to final installation.

During construction, when pipes may be located outside the secured area for welding and/or laying purposes, they shall be in a safe and stable location, secured from movement via wedges and capped to stop any vermin or the public, access internally into the pipes.

11.1.3. Materials

Permanent materials shall fully comply with this Specification and the documents referenced herein. The CONTRACTOR shall prepare and submit CONTRACTORS Certificates for all permanent materials to be included in the works for EMPLOYER's approval and the Material Safety Data Sheet

(MSDS) for all chemicals and drilling materials should also be made available to the CONTRACTOR and EMPLOYER.

The CONTRACTOR shall ensure that chemicals and hydrocarbons are used according to EMPLOYER's accepted environmental practices complete with control measures to mitigate risk.

The CONTRACTOR shall ensure that the drilling fluids and chemicals that have the potential to come into contact with the ground are biodegradable, safe to water bodies and fire resistant.